

Artificial Intelligence Reinforcement Learning

Session 1

What is Artificial Intelligence and why did it develop the way it did?

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ENSISA, Université Haute-Alsace

Course Overview

Course Organization

- ▶ Lectures (CM): 3 sessions, 2 hours each
- ▶ Tutorials (TD): 7 sessions
- ▶ Final exam: 1 session

Course Goal

Understand how Artificial Intelligence evolved from symbolic reasoning and search to sequential decision-making and Reinforcement Learning.

Course Content

- ▶ History and foundations of AI
- ▶ Turing machines and the Turing Test
- ▶ Classical AI and search algorithms
- ▶ Shortest path: Dijkstra and A*
- ▶ Game playing: Minimax and Alpha-Beta
- ▶ Sequential decision-making
- ▶ Introduction to Reinforcement Learning

Main Question

How can a machine reason, search, plan, and eventually learn how to make decisions?

Contents

History of Artificial Intelligence

Alan Turing

The Birth of Artificial Intelligence

The Dartmouth Conference

Symbolic Artificial Intelligence

From Symbolic AI to Modern AI

From Symbolic AI to Modern AI

Intelligent Agents and Problem Solving

A Fundamental Question

"Can machines think?"

A Fundamental Question

"Can machines think?"

This question has fascinated philosophers, mathematicians, and engineers for centuries.

It eventually led to the birth of one of the most influential scientific disciplines:

Artificial Intelligence

Why Artificial Intelligence?

Artificial Intelligence aims to design systems capable of performing tasks that normally require human intelligence.

Examples include:

- ▶ Playing games (Chess, Go)
- ▶ Understanding natural language
- ▶ Computer vision
- ▶ Robotics
- ▶ Autonomous driving
- ▶ Medical diagnosis

Why Artificial Intelligence?

Artificial Intelligence aims to design systems capable of performing tasks that normally require human intelligence.

Examples include:

- ▶ Playing games (Chess, Go)
- ▶ Understanding natural language
- ▶ Computer vision
- ▶ Robotics
- ▶ Autonomous driving
- ▶ Medical diagnosis

Key Observation

Although these applications appear very different, they all require an agent to perceive, reason, and make decisions.

What is Artificial Intelligence?

There is no universally accepted definition of Artificial Intelligence.

Different researchers emphasize different aspects of intelligence:

- ▶ Thinking like humans
- ▶ Acting like humans
- ▶ Thinking rationally
- ▶ Acting rationally

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Throughout this course, we will mainly view AI as:

Working Definition

The study of intelligent agents capable of perceiving their environment and selecting actions to achieve specific goals.

Foundations Before Modern Computing

Long before electronic computers, several civilizations developed ideas that later became essential to Artificial Intelligence.

► Ancient Greece (4th century BC)

- Aristotle established the foundations of formal logic.
- Logical reasoning and deductive inference became the basis of symbolic AI and automated reasoning.

► Golden Age of Baghdad (8th–13th centuries)

- Al-Khwarizmi introduced systematic computational procedures, from which the word *algorithm* originates.
- Engineers such as the Banū Mūsā brothers and Al-Jazari designed sophisticated automated mechanical devices.

► Industrial Revolution (19th century)

- Charles Babbage designed the Analytical Engine, considered the first design for a programmable mechanical computer.
- Ada Lovelace wrote the first published algorithm intended for such a machine and envisioned that computers could manipulate symbols, music, and language—not just numbers.



Al-Khwarizmi (c. 780–850)



Ada Lovelace (1815–1852)

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Alan Turing (1912–1954)

Alan Mathison Turing was a British mathematician, logician, and cryptanalyst, widely regarded as one of the founders of modern computer science.

His major contributions include:

- ▶ Formalizing the notion of computation.
- ▶ Introducing the concept of the Turing Machine.
- ▶ Designing the Universal Turing Machine.
- ▶ Helping break the German Enigma code during World War II.
- ▶ Proposing the famous *Turing Test* for machine intelligence.
- ▶ **Being portrayed by Benedict Cumberbatch in *The Imitation Game*. Watch if you haven't seen it yet !**

Why is Turing important?

Modern computers, computer science, and Artificial Intelligence all build upon concepts introduced by Alan Turing.



A Fundamental Question

Before computers existed, mathematicians faced a surprising question:

What does it mean to compute?

More precisely,

- ▶ Can every mathematical problem be solved by following a sequence of steps?
- ▶ What exactly is an algorithm?
- ▶ Can every algorithm be executed by a machine?

Alan Turing's Idea

In 1936, Alan Turing proposed a simple abstract machine that could perform any algorithmic computation. This model became known as the **Turing Machine**.

A Simple Computational Task

Imagine a very simple robot whose goal is to add one to a number.

Instead of decimal notation, we use **unary notation**, where the number n is represented by n consecutive ones.

Input

☐ ☐ 1 1 1 ☐ ☐

Output

☐ ☐ 1 1 1 1 ☐

Question

How can a machine perform this task by following only simple mechanical instructions?

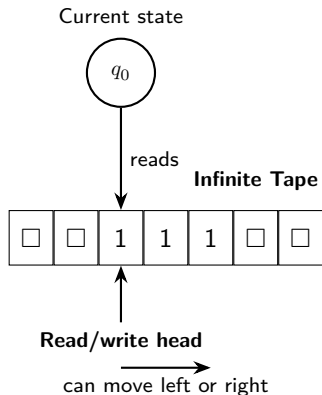
The Turing Machine

Alan Turing imagined a very simple machine that can execute an algorithm. For our example, the machine only needs four simple instructions:

1. Read the current cell.
2. If it contains a 1, move one cell to the right.
3. If it is blank ($\square$), write a 1.
4. Stop.

The machine has only three capabilities:

- ▶ It can read the symbol in the current cell.
- ▶ It can write a new symbol.
- ▶ It can move one cell to the left or to the right.



A Turing Machine at Work

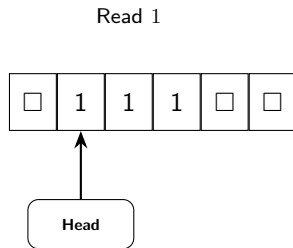
Task: Add one to the unary number 111.

The machine follows two rules:

Rule Book

Symbol read	Action
1	Move one cell to the right
□	Write 1 and halt

The head starts on the first 1.



Step 1: Start

A Turing Machine at Work

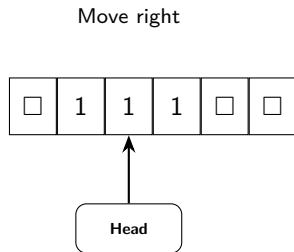
Task: Add one to the unary number 111.

The machine follows two rules:

Rule Book

Symbol read	Action
1	Move one cell to the right
□	Write 1 and halt

The machine reads a 1, so it moves right.



Step 2: Move right

A Turing Machine at Work

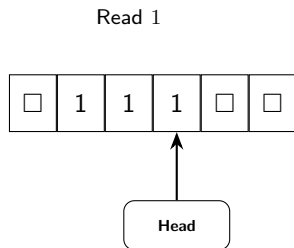
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Step 3: Read the next 1

A Turing Machine at Work

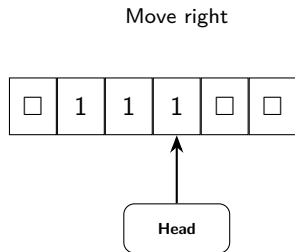
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1	Move one cell to the right
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The machine reads a 1, so it moves right.



Step 4: Move right again

A Turing Machine at Work

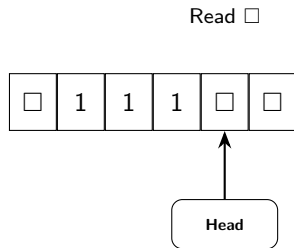
Task: Add one to the unary number 111.

The machine follows two rules:

Rule Book

Symbol read	Action
1	Move one cell to the right
□	Write 1 and halt

The machine reaches a blank cell.



Step 5: Find the blank

A Turing Machine at Work

Task: Add one to the unary number 111.

The machine follows two rules:

Rule Book

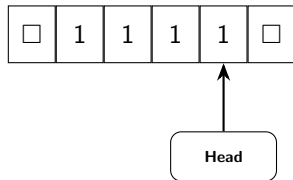
Symbol read	Action
1	Move one cell to the right
□	Write 1 and halt

It writes a 1 and stops.

Result

The input 111 becomes 1111, so the machine has added one.

Write 1 and halt



Step 6: Write and halt

From One Algorithm to Universal Computation

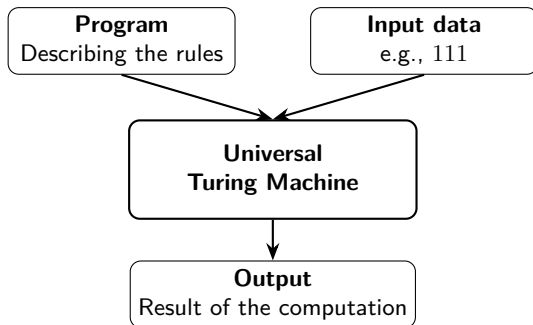
Our first Turing machine was designed for one specific task:

Add one to a unary number

The Limitation

A different task would normally require a different set of rules: Addition, multiplication, sorting, searching, or playing a game **would each require a different Turing machine.**

A standard Turing machine is built to execute one fixed algorithm.



The Big Idea

A **Universal Turing Machine** receives both:

- ▶ a description of an algorithm, and
- ▶ the input on which that algorithm should operate.

It can then simulate any Turing machine, and therefore execute any algorithmic computation that can be expressed by such a machine.

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Can machines think?

By the 1950s, programmable computers were becoming a reality.

The next challenge was no longer:

“Can a machine compute?”

Instead, researchers began asking:

“Can a machine exhibit intelligent behavior?”

A Turning Point

This question marked the beginning of Artificial Intelligence as a scientific field.

The Turing Test

Alan Turing proposed an unexpected idea.

Instead of asking

"Can a machine think?"

he suggested asking a different question:

"Can a machine imitate a human?"

Key Insight

Rather than debating the meaning of *thinking*, Turing proposed evaluating a machine by observing its behavior.

Fun Fact

The title *The Imitation Game* is a reference to Turing's famous thought experiment, the *imitation game*, which later became known as the **Turing Test**.

The Imitation Game

A human judge communicates with two hidden participants using **text only**.

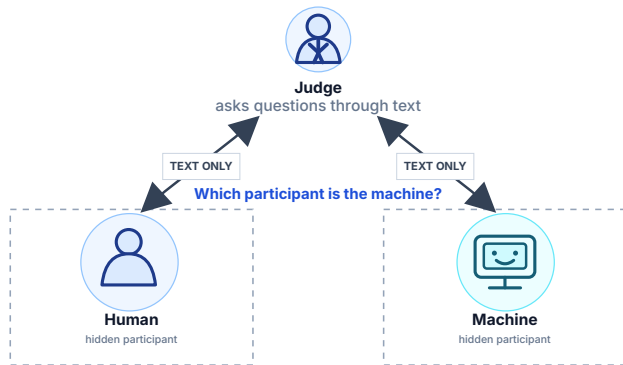
One participant is:

- ▶ a human,
- ▶ a machine.

The judge's goal is simple:

Question

Which participant is the machine?



The Turing Test

If the judge cannot reliably distinguish the machine from the human, the machine is said to have **passed the Turing Test**.

Did the Machine Pass?

Suppose the judge talks to both participants for several minutes.

Question

The judge correctly identifies the machine only about **50% of the time**.

Did the machine pass the Turing Test?

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Yes.

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Did the machine pass the Turing Test?

Answer

Yes.

Why?

The goal is **not** to convince every judge.

A machine passes if a human judge **cannot reliably distinguish** the machine from a human based only on their conversation.

Does the Turing Test Still Matter?

The Turing Test was one of the first attempts to evaluate machine intelligence.

1950

One general test

Human conversation

"Can it imitate a human?"

Today

Many specialized benchmarks

Language, vision, planning, robotics

"Can it solve the task?"

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Human conversation

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Key Idea

The Turing Test remains historically important, but today AI systems are evaluated using many task-specific benchmarks.

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1956: The Birth of Artificial Intelligence

Dartmouth Summer Research Project on Artificial Intelligence

- ▶ Held during the summer of **1956**
- ▶ Brought together researchers interested in machine intelligence
- ▶ Organized primarily by **John McCarthy**

The Proposal

"Can every aspect of learning or any other feature of intelligence be so precisely described that a machine can be made to simulate it?"

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Historic Milestone

The term **Artificial Intelligence (AI)** was introduced, marking the birth of AI as a scientific field.



Figure: Dartmouth College, Summer 1956

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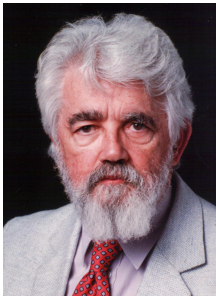
Figure: Dartmouth College, Summer 1956

Historical Note:

Alan Turing is not pictured here. He passed away in **1954**, two years before the Dartmouth Conference. However, his pioneering work on computation and machine intelligence laid the foundations for the field of Artificial Intelligence.

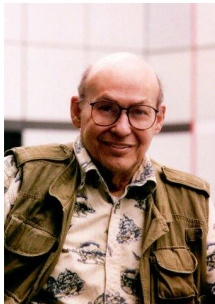
The Pioneers of Artificial Intelligence

The Dartmouth Conference brought together researchers from different disciplines, whose ideas shaped the future of Artificial Intelligence.



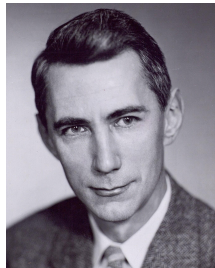
John McCarthy

Coined the term
Artificial Intelligence.



Marvin Minsky

A leading researcher in AI and
cognitive science.



Claude Shannon

Founder of information theory and
an early AI pioneer.



Nathaniel Rochester

IBM researcher and co-organizer of
the Dartmouth workshop.

Legacy of Alan Turing

Although Alan Turing did not attend the Dartmouth Conference, his pioneering work on computation and machine intelligence laid the intellectual foundations on which the field of Artificial Intelligence was built.

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How Did Early AI Work?

How can a machine become intelligent?

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The Early AI Hypothesis

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Knowledge $\rightarrow$ Symbols $\rightarrow$ Reasoning $\rightarrow$ Decision

How Did Early AI Work?

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Researchers believed that intelligence could be achieved by representing knowledge using **symbols** and manipulating those symbols with **logical rules**.

Knowledge → Symbols → Reasoning → Decision

Key Idea

Instead of learning from data, early AI systems relied on human-defined knowledge and logical reasoning.

What is Symbolic Artificial Intelligence?

Definition

Symbolic AI is an approach to artificial intelligence in which knowledge is represented using **symbols** and intelligent behavior is produced by applying **logical rules** to those symbols.

Knowledge Representation

- ▶ Facts
- ▶ Objects
- ▶ Relationships

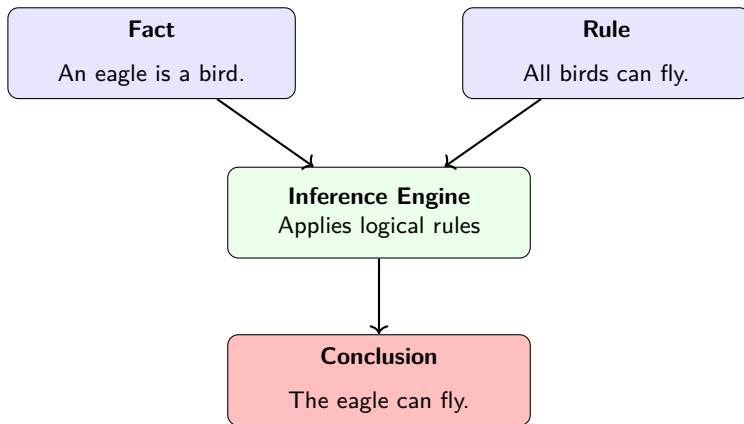
Reasoning

- ▶ Logical inference
- ▶ Rule-based decisions
- ▶ Search and planning

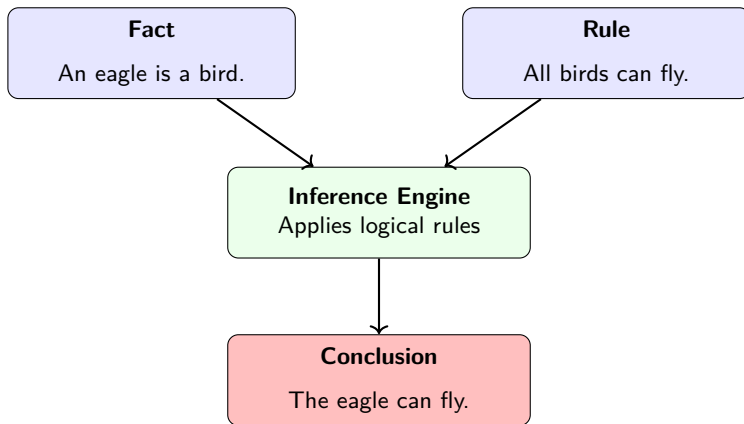
Key Idea

Knowledge is explicitly programmed by humans rather than learned from experience.

Example: Symbolic Reasoning



Example: Symbolic Reasoning



Key Idea

A symbolic AI system combines **facts** with **logical rules** to infer new knowledge. It does not learn from data, it reasons using knowledge provided by humans.

ELIZA (1966)

ELIZA, developed by **Joseph Weizenbaum**, was one of the first conversational AI programs.

Instead of understanding language, ELIZA used simple **pattern-matching rules** to generate responses.

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Example

User: I feel sad.

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Example

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ELIZA: Why do you feel sad?

Key Idea

ELIZA gave the illusion of understanding, but it did not truly understand the conversation.



```
ELIZA Terminal

Welcome to

EEEEEE LL      IIII ZZZZZZ  AAAAA
EE      LL      II       ZZ  AA  AA
EEEEEE LL      II       ZZ  AAAAAA
EE      LL      II       ZZ  AA  AA
EEEEEE LLLLLL  IIII ZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU: █
```

ELIZA (1966)

www.masswerk.at/elizabot/eliza.html

The Legacy of Symbolic AI

Strengths

- ▶ Logical reasoning
- ▶ Explainable decisions
- ▶ Effective for expert systems
- ▶ Foundation of early AI

Limitations

- ▶ Knowledge must be programmed
- ▶ Poor at handling uncertainty
- ▶ Difficult to scale
- ▶ Cannot learn from experience

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The Legacy of Symbolic AI

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What if a machine could learn instead of relying on hand-written rules?

Symbolic AI → Learning-Based AI → Reinforcement Learning

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After the excitement of the 1950s and 1960s...

Many researchers believed that human-level artificial intelligence was just around the corner.

However, several challenges became apparent:

- ▶ Symbolic AI struggled with complex real-world problems.
- ▶ Computers had limited processing power and memory.
- ▶ Expectations greatly exceeded actual capabilities.

The AI Winter



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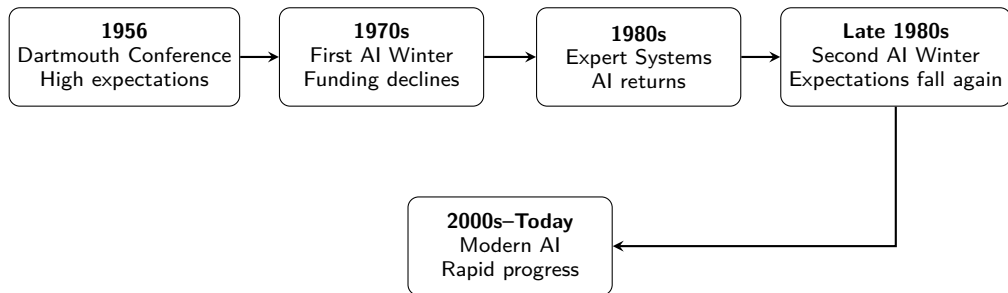
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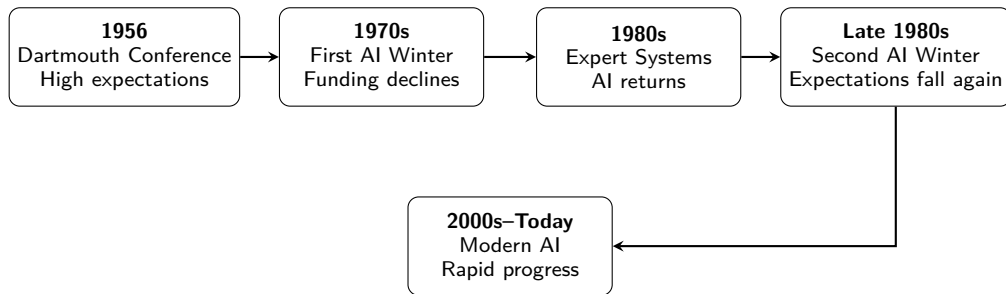
AI Winter

Funding declined, many research projects ended, and enthusiasm for AI decreased.

From AI Winter to AI Revival



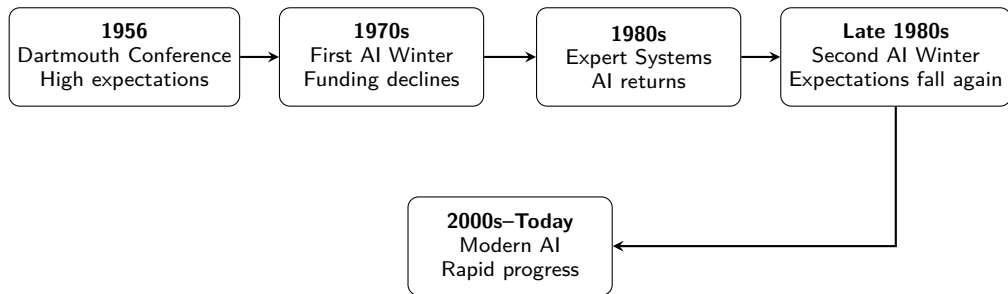
From AI Winter to AI Revival



Why Did AI Return?

- ▶ More powerful computers
- ▶ Large amounts of digital data
- ▶ Improved algorithms
- ▶ Greater investment in AI research

From AI Winter to AI Revival



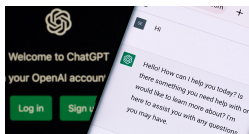
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What can artificial intelligence do today?

Artificial Intelligence Today

Where do you encounter AI in your daily life?



Chatbots



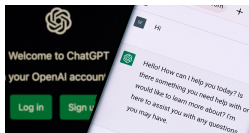
Autonomous Vehicles



Recommendations

Artificial Intelligence Today

Where do you encounter AI in your daily life?



Chatbots



Autonomous Vehicles



Recommendations



Healthcare



Translation



Game Playing

Major Areas of Artificial Intelligence

Problem Solving and Reasoning

- ▶ Search and Planning
- ▶ Knowledge Representation
- ▶ Automated Reasoning

Learning

- ▶ Machine Learning
- ▶ Deep Learning
- ▶ Reinforcement Learning

Games and Adversarial Search

- ▶ Game Playing
- ▶ Minimax
- ▶ Alpha-Beta Pruning

Perception and Interaction

- ▶ Computer Vision
- ▶ Natural Language Processing
- ▶ Robotics and Autonomous Systems

This Course

We will focus on: **Search**, **Game Playing**, and **Reinforcement Learning**.

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What is an Intelligent Agent?

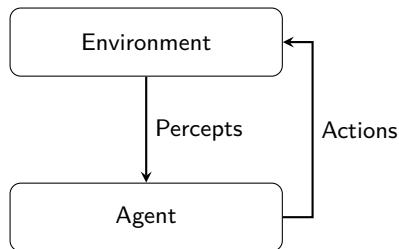
Definition

An **intelligent agent** is an entity that:

- ▶ perceives its environment,
- ▶ makes decisions, and
- ▶ performs actions to achieve a goal.

Examples

- ▶ Self-driving car
- ▶ Chess-playing AI
- ▶ Robot vacuum
- ▶ ChatGPT



Key Idea

AI systems are commonly modeled as **agents** that continuously **perceive**, **decide**, and **act**.

Examples of Intelligent Agents

Agent	Percepts (Inputs)	Actions (Outputs)
Robot Vacuum	Obstacle sensors, dirt sensor, battery level	Move, turn, clean, dock
Chess AI	Current board position	Select next chess move
Self-Driving Car	Cameras, LiDAR, GPS, radar	Steer, accelerate, brake
ChatGPT	User prompt	Generate a text response

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Observation

Although these agents solve very different problems, they all follow the same cycle:

Perceive → Decide → Act

What Makes an Agent Intelligent?

Rational Agent

A **rational agent** chooses the action that is expected to achieve its goal based on:

- ▶ what it has perceived,
- ▶ what it knows about the environment,
- ▶ and the actions available to it.

Examples

- ▶ A robot vacuum chooses the room with the most dirt.
- ▶ A GPS selects the fastest route to your destination.
- ▶ A chess program chooses the move that maximizes its chance of winning.
- ▶ A self-driving car brakes when a pedestrian is detected.

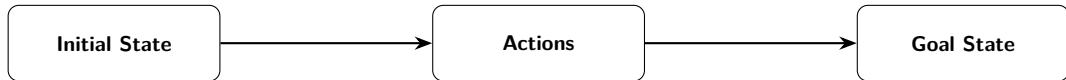
Key Idea

A rational agent does **not** always make the perfect decision. It chooses the **best action available** given its information.

AI as a Problem Solver

Key Idea

Many AI tasks can be viewed as finding a sequence of actions that transforms an **initial state** into a **goal state**.



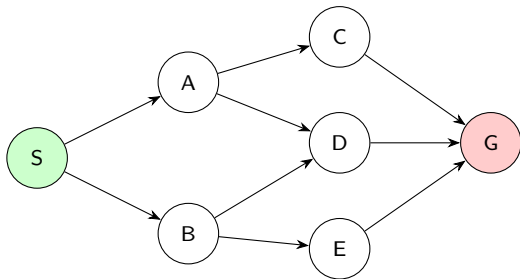
Examples

- ▶ A GPS finds a route from your current location to your destination.
- ▶ A chess program searches for a sequence of winning moves.
- ▶ A robot plans how to reach an object while avoiding obstacles.

State Space

Definition

A **state** represents a possible situation of the problem.
The collection of all possible states is called the **state space**.



Observation

An AI agent must determine **which sequence of actions** leads from the **initial state (S)** to the **goal state (G)**.

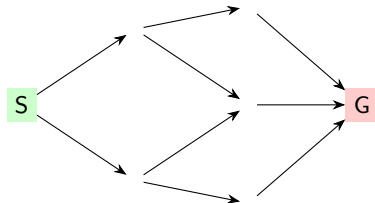
The Challenge of Search

A Simple Question

Given a state space, how should an intelligent agent find a path from the **initial state** to the **goal state**?

Possible Challenges

- ▶ Many possible paths
- ▶ Dead ends
- ▶ Different path costs
- ▶ Limited time and memory



Next Lecture

We will study search algorithms that enable AI agents to efficiently explore state spaces, including **Breadth-First Search (BFS)**, **Depth-First Search (DFS)**, **Dijkstra's Algorithm**, and **A* Search**.